

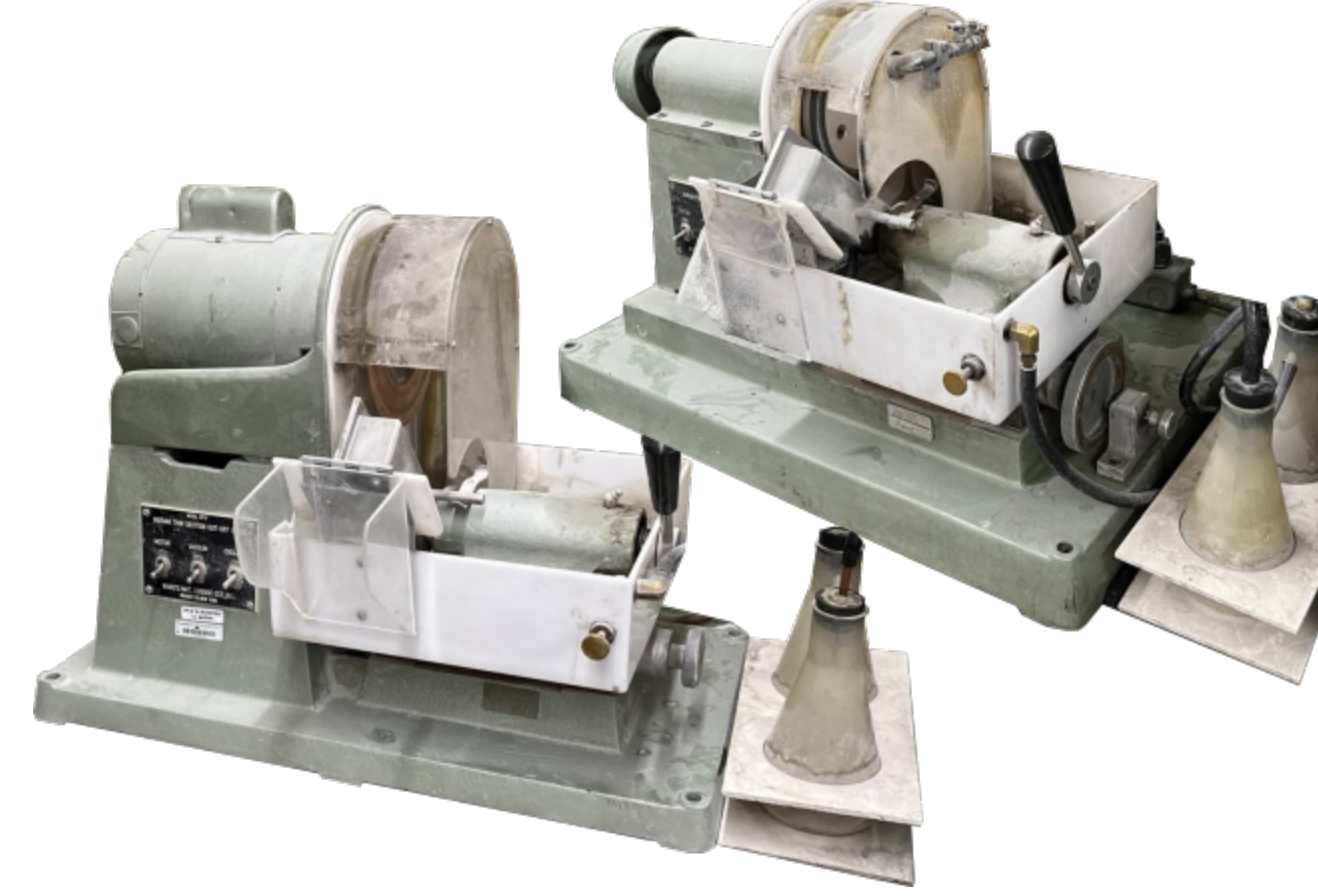


Geosciences sample preparation resources



Water-cooled saw

A wet tile saw, or masonry saw, is a specialized power tool used to cut hard materials like stone, ceramic, and brick. It features a diamond-coated circular blade and uses a constant stream of water to cool the blade, reduce dust, and prevent material fractures. In geology and materials science, this tool is essential for preparing rock, mineral, or ceramic samples with clean, precise cuts for analysis, thin sections, or testing, ensuring structural integrity and accurate examination of internal features.



Thin section saw & grinder

A petrographic thin section saw and grinder are used together to prepare thin rock slices for microscopic analysis. The saw precisely cuts small rock samples, while the grinder thins them to about 30 microns. This process is essential in geology for examining mineral composition, texture, and microstructures in detail.



Low-powered binocular microscope

A low-powered binocular microscope provides magnified, stereoscopic views of samples, allowing detailed observation with both eyes. In geology and materials science, it's commonly used for manual mineral picking with tweezers, helping identify and isolate specific minerals or particles for further analysis or sample preparation.



Oil-cooled saw

An oil-cooled saw is a precision cutting tool designed for sectioning hard or delicate materials, using a thin, diamond-embedded blade lubricated and cooled by circulating oil. Unlike water-cooled systems, the oil reduces friction, heat, and the risk of corrosion, making it ideal for sensitive samples. In geology and materials science, oil-cooled saws are commonly used to cut rocks, minerals, metals, and ceramics for thin section preparation or microstructural analysis, preserving fine details and minimizing damage to the sample's internal structure.



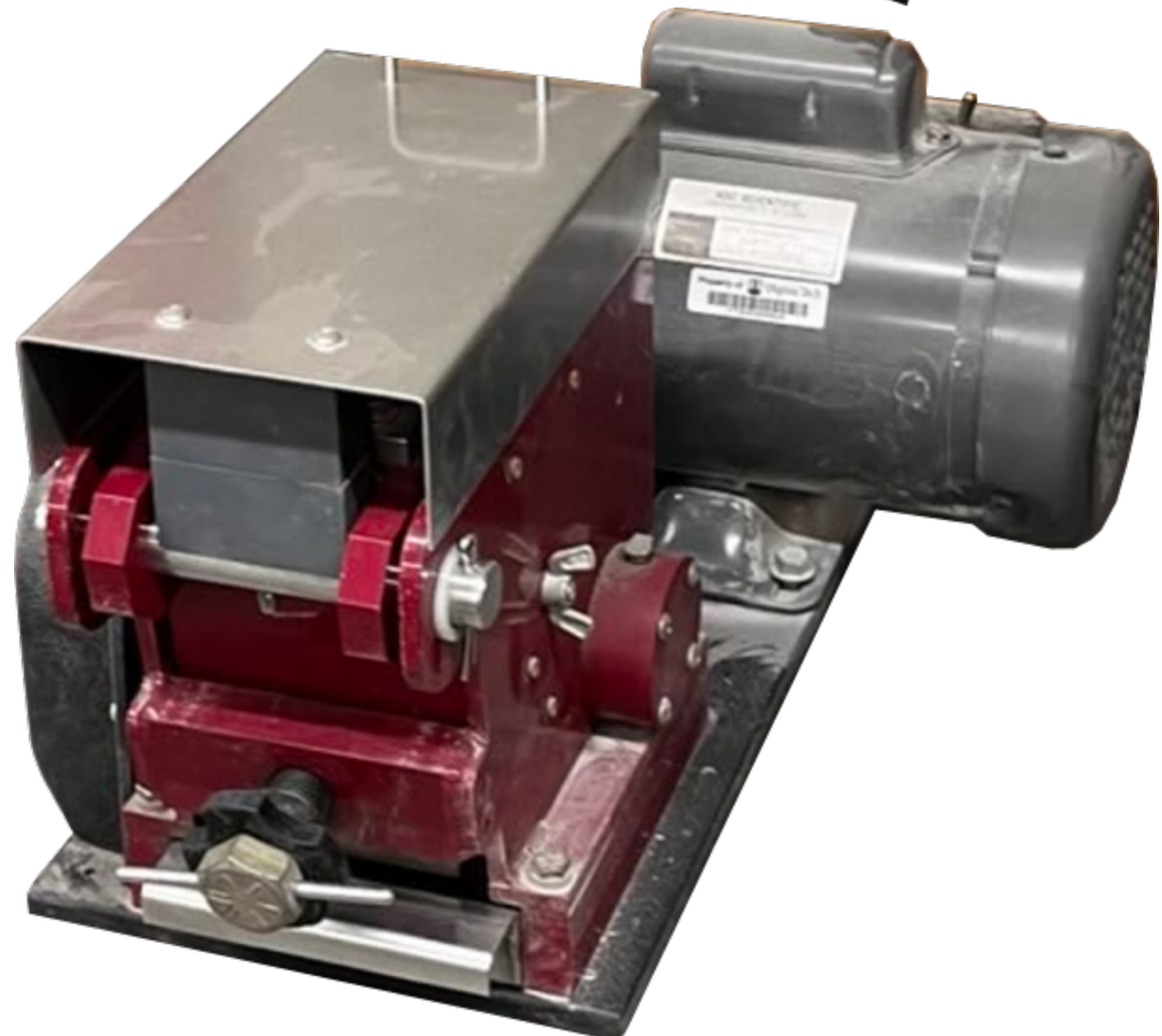
Lapping wheel

A motorized lapping wheel is a precision tool used to polish and flatten materials by rotating a flat, abrasive surface. In geology and materials science, it's essential for preparing smooth, even surfaces on rock, mineral, or metal samples. This helps reveal fine details and microstructures, improving accuracy in microscopic examination and analysis.



Petrographic microscope

A petrographic microscope is a specialized optical microscope designed for examining thin rock sections under polarized light. In geology, it reveals mineral composition, texture, and optical properties, aiding in detailed identification and analysis of rocks and minerals to understand their formation and history.



Jaw crusher

A laboratory jaw crusher is a mechanical device used to crush hard materials like rocks, minerals, and ores into smaller fragments. Common in geology and materials science, it prepares samples for further analysis by applying compressive force between two hardened jaws, ensuring consistent particle sizes for testing or processing.



Flusion furnace

An automated fusion furnace melts samples with flux to create homogeneous glass beads for chemical analysis. Widely used in geology and materials science, it ensures consistent fusion of rocks or minerals, enabling accurate elemental analysis by techniques like XRF, improving precision and reproducibility in geochemical testing.

Pellet press

A manual hydraulic press applies controlled pressure to compress powders into solid pellets. In geology and materials science, it's used to prepare uniform pellets from powdered rock or mineral samples for XRF analysis, ensuring consistent density and shape for accurate and reliable elemental testing.



Carbon evaporation coater

A carbon evaporation coater deposits a thin, uniform carbon layer on sample surfaces using vacuum evaporation. This improves electrical conductivity, essential for imaging non-conductive materials like minerals in electron microscopy. The coating prevents charging and image distortion, enabling clear, high-resolution images. In geology and materials science, it's crucial for detailed microstructural and elemental analysis.



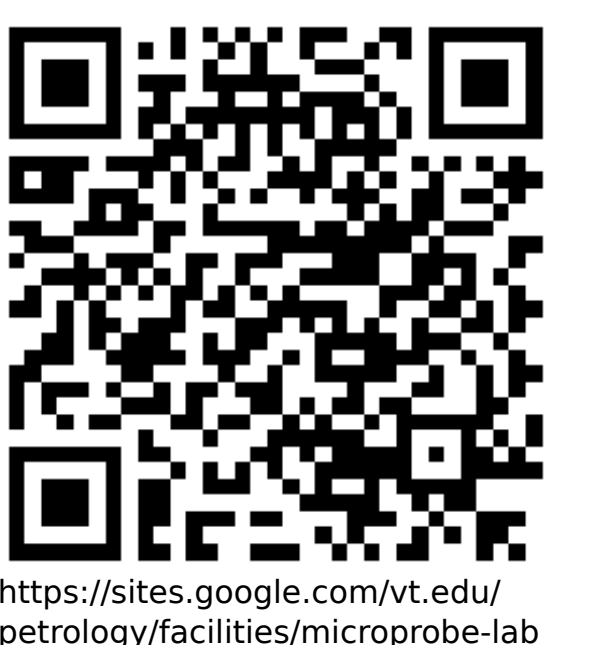
Ball mill

A laboratory ball mill is a device that grinds and mixes materials by shaking and rotating a container flooded with one or more grinding balls. Widely used in geology and materials science, it pulverizes rocks, minerals, and powders into fine, uniform particles. This process is essential for sample preparation, ensuring consistency and homogeneity, which allows for accurate chemical, mineralogical, and structural analysis in research and quality control applications.

Contact

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Rev. 10/21/2025 LRM

Example workflows

